Clovis SFEIR

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Profile

IMT Atlantique engineering student specialising in performance-critical scientific software. Built a finite-difference PDE solver (OpenMP) verified to **2nd-order convergence** against a closed-form solution, a Peng-Robinson/Monte Carlo CO_2 mass-accounting toolkit, an AVX2 pricing engine at **150.9 M options/s**, and C++20 systems benchmarked at nanosecond scale with ASan/UBSan-clean test suites.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*

Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France ***Class rank: 1/85*** *| Software Engineering, Databases, Cybersecurity*

HRT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) — **24th/124** (top 20%); hybrid analytical placer: force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePlAce (1.458); field includes Stanford, Princeton, ETH Zürich.

Projects

Reservoir Diffusivity PDE Solver *C++20, OpenMP, finite-difference*

- Explicit FTCS and implicit Crank-Nicolson finite-difference schemes for the reservoir diffusivity (pressure-transient) equation; verified via grid-refinement convergence study against the closed-form erfc analytical solution – error ratio converges to **4.0** as dx halves (textbook 2nd-order accuracy) across 5 refinements.
- OpenMP-parallelized stencil/assembly loops; measured (not assumed) scaling identifies the Amdahl's-law-bounded sequential Thomas solve and memory-bandwidth-bound behavior below L3 size; 13 test cases, 40,300 assertions; Docker image runs the full suite at build time.

CCUS Injection MRV *Python, Peng-Robinson EOS, Monte Carlo, CUSUM*

- CO_2 density from a Peng-Robinson equation of state with a volume-shift calibrated on one CoolProp point: error $\leq$1.4% away from the critical point, $\approx$6% near it (measured and disclosed); cumulative injected mass with Monte Carlo uncertainty (1,000 draws, $\pm$2.1%).
- CUSUM leak detector on an injectivity model: precision **0.993**, recall **0.965**, detection delay 0.83 h on a labeled synthetic leak; FastAPI service, 53 tests, Docker build verified in CI.

Options Pricing Engine *C++20, AVX2, Monte Carlo, finite differences*

- Black-Scholes, Crank-Nicolson finite differences, Sobol quasi-random Monte Carlo and SABR calibration, with AVX2 SIMD kernels and a Google Benchmark suite.
- Measured **150.9 M options/s** for AVX2 Black-Scholes (i7-12700K, GCC 12, `-O3 -march=native`), reproducible from the benchmark suite.

Crude Oil Spread Trading Engine *C++20, stochastic processes, order book*

- Models refining margins (3:2:1 crack spread, generalized N:M:K) and mean-reverting inter-crude spreads via an Ornstein-Uhlenbeck process (Euler-Maruyama) with a rolling z-score signal, matched by a price-time-priority order book.
- OU stationary distribution verified against theory (500K-step sim); crack-spread formula checked to 10^{-9}; order-book submit at **720 ns** (non-crossing)/**1.24 µs** (crossing); 58 assertions/21 tests, ASan+UBSan clean.

Work Experience

Infotel *Nantes, France*
AI Engineer Apprentice – Lab IA *Sep 2025 – Jun 2028*

- Built a Rust-based autonomous software-delivery harness that takes a complete specification through architecture synthesis (subsystems, interface contracts), epic/slice decomposition, and TDD-first generation to a tested, deployable application end-to-end; automated compile+test quality gates with iterative self-repair, independent code-review pass, and cross-slice conflict detection before delivery; 928 Rust tests; validated end-to-end producing a fully working, 100%-test-passing application directly from a single specification.

Excelia Group *La Rochelle, France*
Software Developer Intern / AI Software Engineer Intern *Two internships: Jan–Mar 2025 & May–Jul 2024*

- RAG pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking): **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + asyncio parallelisation; Prometheus/Grafana, Docker, GCP.
- Real-time interview simulator (Next.js/React, GPT-4o, ElevenLabs TTS): streaming responses, scenario-branching state machine, A/B prompt-scoring harness.

Technical Skills

Perf / Systems	OpenMP, cache-aware layout, acquire/release atomics, ICS/SCADA protocol parsing (Modbus, IEC-104)
Languages	C++20, Python, Rust, TypeScript, SQL
Numerics	finite-difference PDE (FTCS, Crank-Nicolson), Monte Carlo/Sobol, Peng-Robinson EOS, Ornstein-Uhlenbeck, CUSUM, BM25
Infrastructure	Docker, tokio async, Prometheus/Grafana, GitHub Actions CI, GCP, Redis, ZeroMQ
Frameworks	FastAPI, Flask, Node.js/Express, React/Next.js, REST/gRPC